

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2015

Subject: 392131 Physics I

Section: 15 - 16

Date: 15 October 2015

Time: 10.00 - 12.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken inside the examination room.
3. Textbooks are **NOT** allowed.
4. Express your answer in English. Other languages will be discarded.
5. All calculators are **NOT** allowed.
6. Electronic Communication devices are **NOT** allowed in the examination room.
7. The examination has 9 pages (including this page), 21 questions and a total score of 100 points.
8. Write all your answers on this exam sheet.

Name _____ ID _____ seat No. _____ 392131/2

1. Write down in blank table for 7 base quantity units in SI Unit (6 points)

Base quantity	Name	symbol
	meter	
Mass	kilogram	
Time		S
Electric current		
		K
Amount of substance		
Luminous intensity		

2. Power (P) can be written in terms of the work (W) and time (t) as $P = \frac{W}{t}$. If Unit of work is Joule (J) and unit of time is second (s). Show that unit of power in term of SI Unit. (Give $W = \vec{F} \cdot \vec{s}$, Unit of F is $\text{kg} \cdot \text{m} / \text{s}^2$ and unit of S is meter) (2 points)

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3. How many significant figures are there in (3 points)

- | | | | |
|---------------------------|-------|-----------------------|-------|
| a) 0.10070 | | b) 4.00×10^9 | |
| c) 100.70 | | d) 0.0030 | |
| e) 1.019×10^{-6} | | f) 4.87 ± 0.06 | |

4. Round-off 9,191919 . . . to 2 decimal places (1 point)

5. Converse this number to prefixes of Base Units (3 points)

40000 m can be written as _____ km

0.0001 g is the same as _____ mg

$2.5 \times 10^6 \text{ N}$ can be written as _____ GN

250000 A can be written as _____ kA or _____ MA

0,000000075 s can be written as _____ ns

$3 \times 10^{-7} \text{ mol}$ can be rewritten as _____ mol, which is the same as _____ $\mu \text{ mol}$

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6. The Concorde is a type of aeroplane that flies very fast. The top speed of the Concorde is 844 km/h. Convert the Concorde's top speed to m/s. (2 points)

7. What are random errors? How can we minimize these errors? (2 points)

8. Use the rules for significant figures to find the answer to the addition problem (1 point)

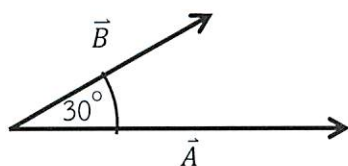
$$0.043 + 15.0 + 117 + 4.301 = \dots\dots\dots$$

9. The length, breadth and thickness of a rectangular plate are 4.0 m, 10.005 m and 12.01 cm respectively. Find the total area and volume of the plate to correct significant figures. (3 points)

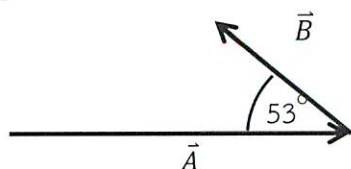
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10. Find Subtraction and Addition of Vectors. (Magnitude and direction with positive x-axis) Give magnitude of $\vec{A}$ is 20 unit and magnitude of $\vec{B}$ is 15 unit. (6 points)

10.1

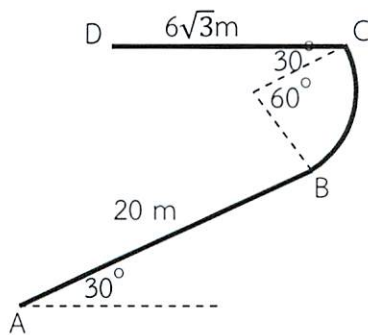


10.2



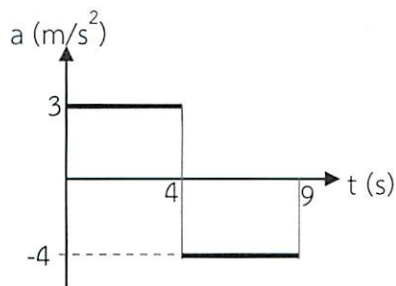
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11. A train moves from A through B and C to D as shown in figure. Find average speed and average velocity if total time in this travel is 20 seconds and path in B to C is part of circle radius 2 m. (In term unit m/s) (10 points)



12. A car moves in uniform acceleration along a straight line show the graph of acceleration versus time of car in Figure. If this car starts in 10 m/s out of start point and far from starting point 15 m. Find (9 points)

- (a) The average velocity for the total trip,
 (b) Final velocity and position at 9 second. (Position refer to starting point)
 (c) The instantaneous velocity at $t = 4$ s.



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13. On a roller coaster ride at an amusement park, a car travels from 76 m/s to 56 m/s in 4.0 seconds.

(a) What is the car's acceleration? (2 points)

(b) If the car continued this acceleration, how fast would it be traveling after 150 m? (3 points)

14. A man running north for 90.0 minutes at 90.0 km/h and then stops for drinking water 10.0 minutes.

And then he continues running in west, traveling 300 km in 4.00 h. (6 points)

(a) What is his total displacement and total distance?

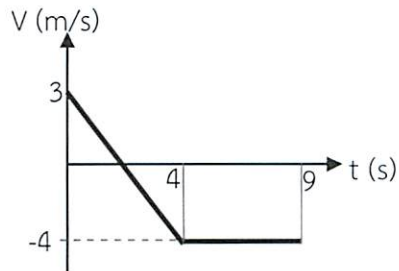
(b) What is his average velocity and average speed?

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15. According to an article in Business Week, the 2008 Corvette can go from zero to 72 km/h in just 4 seconds. Assuming acceleration is uniform, then how much track length did it take to accomplish this? (3 points)

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16. From velocity versus time, drawing the graph acceleration versus time and displacement versus time (start with position -3 m from reference point) (6 points)



17. Mama thrown vertical upward the ball from 10 m/s. Draw graph acceleration versus time, velocity versus time and displacement versus time of this. (6 points)

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18. A bullet is fired at Wonder Woman. The bullet leaves the gun's muzzle at 1000 m/s. Wonder Woman is standing 10 meters in front of the bullet. The instant the bullet is fired Wonder Woman begins to move her hand to block the bullet. Her hand starts from rest. She has to move her hand 1.00 meters to block the bullet. **(10 points)**
- a. When the bullet is in the air it will slow down at a rate 30 m/s^2 . How long did it take for the bullet to reach Wonder Woman?
 - b. How fast was the bullet traveling when Wonder Woman deflected it?
 - c. What was the average speed that Wonder Woman moved her hands to deflect the bullet?
 - d. What was the final speed of Wonder Woman's hand when she deflected the bullet?
 - e. What was the acceleration of her hand?
 - f. Wonder Woman stopped her hand in 0.3 m. What is the acceleration of her hand now?

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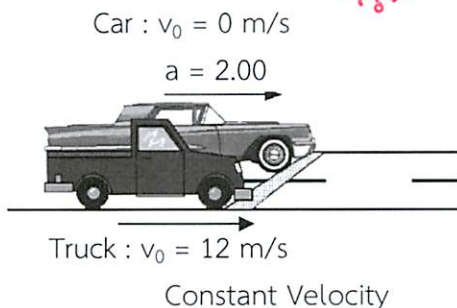
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19. A ball is thrown vertically upward with a speed of 5 m/s. (4 points)

- (a) How high does it rise?
- (b) How long does it take to reach its highest point?

20. A car is at rest at a stop light. The moment the light turns green a truck rolls up the line with a constant velocity of 12 m/s. At the instant the truck is next to the car; the car begins to accelerate as shown. (6 points)

- (a) How much time does it take for the car to catch up to the truck?
- (b) How much distance is covered when the from the start line to when the car catches up to the truck?
- (c) What is the velocity of the car when it catches up to the truck?



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21. A rocket moves straight upward, starting from rest with an acceleration of 72 km/h . It runs out of fuel at the end of 20.00 s and continues to coast upward, reaching a maximum height before falling back to Earth. Find the maximum height the rocket reaches and total time before reaching to ground. (6 points)

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Dr. Dusit Ngamrungrroj